

CHAPTER FOUR

CONCLUSION

The geochemical analysis of sediments in the study area reveals important information about their origin and the processes that have shaped them. The sediments show intense weathering of the source area, indicating significant chemical breakdown of minerals. They have a mature polycyclic quartz provenance, suggesting multiple cycles of erosion and deposition from felsic to intermediate igneous rocks. Hydraulic sorting processes have played a role in segregating different mineral components, with sediments primarily derived from a recycled sedimentary provenance. The sediments are moderately to highly mature and have originated from both weakly and intensively weathering. These findings, along with discriminate diagrams related to clastic sediment, suggest that the sediment was deposited in an active continental to passive continental margin. The mature weathered sediments were derived from the source area after collation, upliftment, and erosion, and were sourced from the Himalaya and deposited in the basin.